Solution Architecture

Date	15 March 2024
Team ID	SWTID-2026-4536
Project Title	LegalEase - AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

Streamlit Web Application

- User Input Interface
- Document Type Selection
- Parties, Terms & Effective Date
- Document Preview
- Edit Document
- Download TXT / DOCX / PDF

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API Gateway / Load Balancer

FastAPI Server

- /generate API Endpoint
- Handles API Requests
- Connects Frontend with AI Core

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Auth Service

Secure Environment Configuration

- .env
- Gemini API Key
- Secure Handling of Sensitive Data

Core Logic Service

LegalEase AI Core

- Gemini 1.5 Pro
- Legal Document Generation
- Structured Prompt Processing
- Document Formatting
- Editable Document Generation

External API

Integrations

Google Gemini Generative AI API

- AI-powered legal content generation
- Gemini generate_content integration

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Data / Storage Layer

Document & File Storage

- Generated TXT Documents
- Generated DOCX Documents
- Generated PDF Documents
- Logos and Formatting Assets

Component Description Table

Component Name	**Description / Role in Architecture**	**Technologies Used**
Presentation Layer	Provides the user interface for entering document type, parties, terms and effective dates, previewing, editing and downloading generated legal documents.	Streamlit

API Gateway	Receives user requests from the frontend and sends document-generation requests to the backend processing layer.	FastAPI
Microservices	Handles legal document generation, AI integration, document formatting, preview generation and export functionality.	Python, Gemini 1.5 Pro, python-docx, FPDF
Database	Stores/generated documents and related application resources required for document generation and export.	TXT, DOCX, PDF files